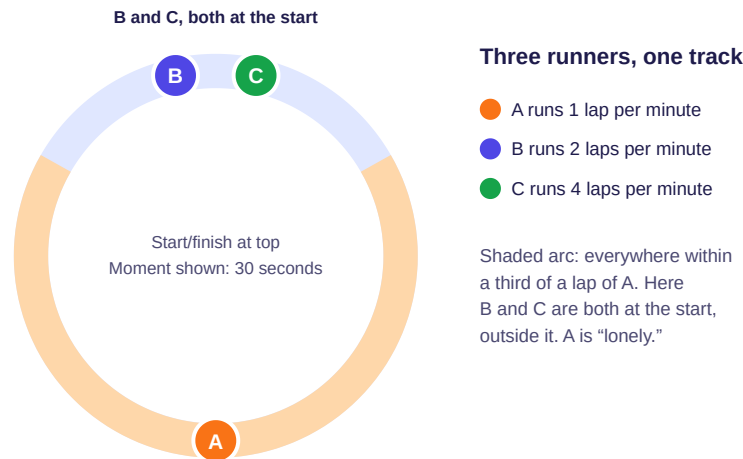


Lonely Runner Conjecture

Runners start together on a round track, all at different speeds. Will every runner, at some moment, be far from everyone else?



After 30 seconds, B and C are back at the start while A is halfway around. A is lonely.

TRY IT

2 runners: "far" means half a lap

3 runners: a third of a lap

4 runners: a quarter of a lap

Speeds 1, 2, 4 laps a minute: at 30 seconds, A is lonely

WHAT MATHEMATICIANS KNOW

- It was first posed in the late 1960s and 1970s, in the study of how lines of sight get blocked, and later renamed the lonely runner conjecture.
- It has been proved for up to seven runners (2008).
- The distance can't be made any bigger: some speeds make runners exactly that far apart and no farther.

WHY IT'S STILL UNSOLVED

Each new runner adds another speed that can get in the way, and the ways runners bunch up get complicated fast. The methods that work for seven runners don't stretch to every number of runners.

TRY THIS WITH STUDENTS

- Act it out: three students walk a circle at one, two, and four steps per clap. Freeze! Who's lonely?
- Connect to rates: where is each runner after half a minute? A third of a minute? It's fractions of a lap in disguise.
- Ask: With two runners, when are they farthest apart? Can you predict it from their speeds?

WORDS TO KNOW

Lap one full trip around the track

Rate distance per unit of time, like laps per minute